

The planet Mercury takes the least time to complete one rotation around the sun. The time taken by all eight planets to complete one revolution around the sun is as follows –

Mercury	-	88 Days
Venus	-	224.7 Days
Earth	-	365.2 Days
Mars	-	687 Days
Jupiter	-	11.86 Years
Saturn	-	29.4 Years
Uranus	-	84.07 Years
Neptune	-	164.9 Years

3. Which of the following pairs of planets is without satellites?

- (a) Venus and Mars
- (b) Mercury and Mars
- (c) Earth and Jupiter
- (d) Mercury and Venus

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Ans. (d)

Venus and Mercury are the only planets in the solar system that do not have any natural satellites or moons, while Earth has 1 moon, Mars-2 moons, Jupiter-95 moons, Saturn-146 moons, Uranus- 27 moons and Neptune has 14 moons.

4. Consider the following statements :

1. The albedo of an object determines its visual brightness when viewed with reflected light.
2. The albedo of Mercury is much greater than the albedo of the Earth.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

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Ans. (a)

The term albedo is commonly used to describe the reflective properties of planets Satellites and asteroids Planet. Mercury receives greater sunlight in comparison to Earth but its albedo is less than that of Earth. The albedo of Mercury is approx 0.1 while the albedo of Earth, it is approx. 0.3. Thus, only the first statement is true that is why option (a) is correct.

v. The Venus

*Venus is also known as the **twin sister** of planet Earth. Venus is the **brightest planet** in the solar system. It is also known as the **Morning Star** and **Evening Star**. The distance between the Sun and Venus is appx. 108.2 million km. It is the second closest planet to our sun. It takes Venus appx. 225 (224.7) days to complete its one revolution. It is the hottest planet in the solar system with an average temperature of 464°C (737K).

*The temperature of the planets in the solar system is as follows –

Mercury	=	167 °C,
Venus	=	464 °C
Earth	=	15 °C,
Mars	=	-65 °C,
Jupiter	=	-110 °C,
Saturn	=	-140 °C,
Uranus	=	-195 °C,
Neptune	=	-200 °C.

*Every planet in our solar system except for Venus and Uranus rotates **clockwise** i.e., from east to west.

*Some evidence of atmosphere is found on the planet Venus.

The atmosphere of Venus mainly comprises of carbon-dioxide (96.5% CO₂), with average temperatures increasing above 400°C. That is why the planet experiences a pressure cooker like condition. Hence because of the above conditions sustenance of any life form becomes next to impossible on this planet. *Just like Mercury & Venus too has **no natural satellite** of its own. *In the year 1989 **Magellan spacecraft** was sent to Venus on a deep space mission for the next 4 years i.e., from 1990-1994. It observed the planet Venus and its features. The satellite was named after **Ferdinand Magellan**, a famous Portuguese explorer of the 16th century.

1. The hottest planet of the solar system is –

- (a) Mercury
- (b) Venus
- (c) Mars
- (d) Earth

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Ans. (b)

Average temperature of all planets in our Solar System is as follows –

Mercury	=	167° C
Venus	=	464° C
Earth	=	15° C
Mars	=	- 65°C
Jupiter	=	-110° C
Saturn	=	-140° C
Uranus	=	-195° C
Neptune	=	-200° C

Thus, Venus is the hottest planet in our Solar System and Neptune is the coldest planet.